Three Directions, and an Objection to the Question

what to actually do with `hecke-in-q-bool.pdf`

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Abstract

`hecke-in-q-bool.pdf` establishes that `q-bool` and `lascoux` contain, unnamed, a number of standard objects of the Langlands programme. Every one of those identifications is classical. This document argues that the obvious continuations — building the affine flag container, chasing the dual group through Satake — are therefore re-derivations rather than research, and that the one place where the container formalism has a feature the Hecke literature lacks is the shape/position asymmetry that `q-container.pdf` Theorem B records as an obstruction and that §2 here reinterprets as a description of a cell decomposition. Three directions follow from that reading. **A** (§3): Hecke correspondences are spans, not maps, so the right container-theoretic home for them is a bicomodule; RSK wants the same missing structure, which joins the two repositories at the level of the open problem rather than the theorem. Highest ceiling, gated on an unobtained paper. **B** (§4): expander codes, joining `error-correcting.pdf` to the Ramanujan-graph face of the Hecke operator — tractable, applied, lower ceiling. **C** (§5): a paving theorem characterising q-containers, which would give `q-container.pdf` the central theorem it currently lacks and explain its own refuted conjecture. §6 says what not to do, §7 gives a sequencing with kill criteria, and §8 gives honest priors.

1 The objection: recognition is not research

The mishnah is that `hecke-in-q-bool.pdf` opens a programme. Ten links, four verifications, a list of next steps.

The objection is that all sixteen rows of its status table are marked **classical** or **open-and-not-attempted**, and that the classical ones are classical by a wide margin: Iwahori 1964, Lascoux–Schützenberger 1982, Kostant–Kumar 1986, Macdonald's Chapter V, Bernstein, Cherednik, Mirković–Vilonen. What the document does is *recognition*. It tells you where you already were. It produces nothing.

The five continuations listed in its *mai nafka minah* inherit that defect. Building the affine flag container terminates in a theorem of Bernstein's. Chasing Satake terminates in Macdonald's Chapter V. Deciding whether the t-adic grading induces Macdonald's q terminates in Cherednik. Each is a re-derivation in container notation of an object that does not need a new presentation, and `lascoux/lenses-in-my-thesis2.pdf` already contains the correct instinct about this in its box headed *what the container language does not buy*.

The resolution is to stop asking which link to chase and ask instead: where does the container formalism have a feature that the Hecke literature lacks? There is exactly one, and `q-bool` has already proved a theorem about it.

2 The asymmetry is a description, not an obstruction

`q-container.pdf` Theorem B states that linearising the shapes destroys the theory: a q-container must keep its shapes a bare set while its positions become $\mathbb{F}_q$-spaces.

`data-control.pdf` elevates this to a principle and argues, in its §"Why the distinction cannot be repaired away", that the set-ness of shapes is not a limitation awaiting a generalisation.

Both documents treat this as an obstruction discovered. Read against §3 of `hecke-in-q-bool.pdf`, it is instead a description arrived at.

Observation 2.1. A Bruhat stratification of a flag variety is exactly a *set* of cells, indexed by the Weyl group W, each of which is an *affine space* $\mathbb{F}_q^{\ell(w)}$. Shapes bare, positions linear. The reason shapes cannot be linearised is that W is not a variety: there is no Weyl-group-shaped vector space for them to become.

This is worth stating carefully because it changes what the formalism is *for*. If Observation 2.1 is right in general, then a q-container is not a deformation of a data type. It is a presentation of a space with an affine paving, and the counting series of `q-container.pdf` §6 is a Poincaré polynomial. All three directions below are consequences of taking that seriously; §5 is the attempt to prove it.

3 Direction A: Hecke correspondences as bicomodules

3.1 The negative result that motivates it

`hecke-in-q-bool.pdf` §10 records, among the things that do not link, that the Hecke correspondence is not a container morphism, because both legs move shapes. That negative is the interesting part.

A container morphism (u, f) is a *map*: u dispatches forward on shapes, f routes backward on positions. A Hecke operator is a *span*:

$$\mathrm{Fl}_n(\mathbb{F}_q) \xleftarrow{p_1} Z_i := \mathrm{Fl}_n \times_{\mathrm{Fl}_n^{(i)}} \mathrm{Fl}_n \xrightarrow{p_2} \mathrm{Fl}_n(\mathbb{F}_q), \qquad Z_i = \{(F, F') : \pi_i F = \pi_i F'\},$$

and the algebra structure is composition of spans by pullback — which is convolution, which is why $T_i T_j$ means anything at all. Spans of G-sets are G-equivariant profunctors; the Hecke algebra is an endomorphism algebra in that bicategory.

So the missing notion is not a better kind of map. It is the layer above maps.

3.2 What that layer is in this formalism

Comonoids in **Poly** are small categories (Ahman–Uustalu for directed containers; Garner), and bicomodules between polynomial comonoids are the profunctor-like layer above them, specialising to spans when the comonoids are discrete. The formalism therefore already has the right shape of object. What it does not have is the $\mathbb{F}_q$-linearised version, which is precisely the `q-bool` setting.

Definition 3.1. Provisional. A *q-bicomodule* from $S \triangleleft P$ to $T \triangleleft Q$ consists of a set R of *correspondence shapes* with maps $\sigma : R \to S$ and $\tau : R \to T$, together with, for each $r \in R$, a linear relation

$$\rho_r \subseteq P(\sigma r) \oplus Q(\tau r),$$

composed by pullback on shapes and composition of linear relations on positions, with multiplicities given by counting fibres.

The move from *map* to *relation* on the position plane is the whole proposal, and it is forced: a Hecke structure constant is a count of intermediate objects, which a function cannot record and a relation can.

3.3 What would count as success

Three milestones, in order, each falsifiable and the first two cheap.

Milestone 1. The Hecke span Z_i of §3 is a q-bicomodule in the sense of Definition 3.1, with R the set of pairs (w, w'), $w' \in \{w, ws_i\}$, and ρ the incidence of Schubert cells. One week.

Milestone 2. Composition of q-bicomodules reproduces

$$T_i^2 \;=\; (q-1)T_i + q\,I.$$

This is the sharp test, and it is decidable against `code/verify.py` V1, which already has the matrices for $(n, q) \in \{(2,2),(2,3),(3,2),(3,3),(4,2)\}$. If the composition law of Definition 3.1 does not produce the q in the second term, the definition is wrong and should be changed rather than defended.

Milestone 3. RSK is a q-bicomodule. `lascoux-container.pdf` §"What RSK becomes" already conjectures that RSK is a bicomodule between the two comonoids, and lists it as open. If Milestones 1–3 all land, the two repositories share not a theorem but an open problem, which is a stronger form of connection: one construction closes both.

Beyond that, the ambitious end: whether the monoidal structure on q-bicomodules reproduces the Satake tensor structure, hence $\mathrm{Rep}({}^L G)$ as structure on a polynomial comonoid. That is the payoff and it is not a milestone, because nothing about it is currently plannable.

3.4 The hazard, which is internal as well as external

Hazard 3.2. Composition in Definition 3.1 requires the pullback on the shape plane to interact coherently with relation-composition on the position plane — that is, a distributive law between the two planes of `data-control.pdf`. The LICS 2024 no-go theorems on directed-container distributive laws bear directly on this. They are listed in `lascoux/papers/INDEX.md` as *paywalled and not obtained*.

This hazard is not merely a missing citation. `data-control.pdf` argues at length that the data/control distinction cannot be repaired away; a distributive law is exactly a repair; so the document's own central principle predicts that the distributive law may fail. If it does, Direction A is dead, and the failure is informative rather than merely disappointing — it would say that Hecke convolution is not expressible in this formalism and would explain why.

Obtain that paper before committing a month. It is an afternoon, and it gates everything above.

4 Direction B: expander codes

The tractable direction, and the only one with applications.

Three facts already in hand, from three different documents:

(i) `error-correcting.pdf`: a container morphism over $\mathbb{F}_q$ dispatches on a shape and then routes data *without inspecting it*; the routing is a matrix, so the morphism is a linear code, composition is matrix product, and rank is monotone under composition.

(ii) `hecke-in-q-bool.pdf` §2: the Hecke operator for GL_2 is the adjacency operator of the $(q+1)$-regular Bruhat–Tits tree, and Deligne's bound $|a_p| \le 2p^{(k-1)/2}$ is the optimal spectral gap — which is where Ramanujan graphs come from (Lubotzky–Phillips–Sarnak, by arithmetic quotient).

(iii) Sipser–Spielman expander codes are built from a d-regular expander plus a small inner code, and decode by a *local* flip rule.

Question 4.1. Is a Sipser–Spielman code a container morphism whose shape set is the vertex set of an expander and whose position spaces carry the inner code? If so the expansion parameter is a property of the *shape* plane and the inner code a property of the *position* plane, and locality of decoding is the statement that routing does not inspect.

That last clause is why this direction is worth doing rather than merely possible. `data-control.pdf`'s distinction is decorative in most of its instances; in a locally-decodable code it does real work, because "the decoder looks at the graph and not at the data" is a correctness property, not a description.

Milestone 4. Express the Tanner / Sipser–Spielman construction as a composite of container morphisms, and check that the rank-monotonicity theorem of `error-correcting.pdf` yields the standard rate bound $R \geq 1 - 2(1 - r)$ for inner rate r. If the formalism cannot recover a bound it should be able to recover, it is not buying anything here either.

Milestone 5. Exhibit an LPS Ramanujan graph as the shape plane of such a container and read the spectral gap off the Hecke operator of `hecke-in-q-bool.pdf` §2 rather than importing it. This is the join: the same operator, in the same q, serving a coding-theoretic purpose.

Ceiling: modest. The coding theory is well-trodden and the container reading is unlikely to improve a bound. Floor: high — the milestones are achievable and the write-up is publishable as an exposition even if nothing is new.

5 Direction C: the paving theorem

Self-contained, improves an existing document rather than starting a new one, and is the direct attempt to prove Observation 2.1.

Conjecture 5.1. *A q-container with finite shape set "is" a finite-type $\mathbb{F}_q$-scheme with an affine paving indexed by that set: shapes index strata, positions are the strata, and the counting series of `q-container.pdf` §6 is the Poincaré polynomial of the paving evaluated at the field size.*

If this is right it does three things at once for `q-container.pdf`.

1. It *explains* Theorem B instead of recording it: shapes cannot linearise because the index set of a paving is not itself a variety.
2. It explains the refuted Conjecture C in the same breath — $\mathbf{Vect}_{\mathbb{F}_q}$ being self-dual is a statement about the position plane, and the failure of a second derivative is a statement that the shape plane has no dual.
3. It supplies the Targets-table row for q-Pascal for free, since the Grassmannian container ought to have counting series $\begin{bmatrix} n \\ k \end{bmatrix}_q$ by the Schubert paving of $\mathrm{Gr}(k, n)$, and q-Pascal is then the cell recursion rather than an identity to be verified.

Milestone 6. Compute the Grassmannian container — shapes the partitions in a $k \times (n - k)$ box, positions $\mathbb{F}_q^{|\lambda|}$ — confirm the counting series is $\begin{bmatrix} n \\ k \end{bmatrix}_q$, and derive q-Pascal from the cell recursion. This is a day, and it is the smallest possible test of Conjecture 5.1.

Milestone 7. State the correspondence in both directions and find the first counterexample. A q-container is bare data; a paved scheme has a topology; so the correspondence is at best an equivalence onto a subcategory, and locating the boundary is the theorem.

Ceiling: modest but real — a characterisation theorem is a better centre for `q-container.pdf` than a list of verified propositions. Probability of something true and provable: high.

6 What not to do

don't Build the affine flag container.
This is `hecke-in-q-bool.pdf` Question 5.4 and it is my own suggestion; I am withdrawing it as a priority. It is the most obvious next step and the least rewarding: a long construction whose endpoint is Bernstein's description of the centre, and whose novelty is notational.

don't Chase Macdonald's second parameter as a construction.
Claim 7.2 of that document is very likely right, and being right costs nothing to record and everything to re-derive.

gated Read Macdonald Chapter V.
Two days, and it is de-risking rather than direction: it decides whether §6 of that document stands on anything. Worth doing, not worth planning around.

7 Sequencing, with kill criteria

	step	cost	kill criterion
1	Obtain the LICS 2024 no-go paper	an afternoon	—
2	Macdonald Ch. V against `q-container` §9	two days	—
3	Direction C, Milestone 6 (Grassmannian)	a day	counting series $\neq \begin{bmatrix} n \\ k \end{bmatrix}_q$
4	Direction C, Milestone 7 (the boundary)	two weeks	no clean subcategory after two weeks
5	Direction A, Milestones 1–2	two weeks	composition does not give $(q-1)T_i + q$
6	Direction A, Milestone 3 (RSK)	a month	the distributive law of Hazard 3.2 fails
7	Direction B if A dies	—	—

Step 1 gates steps 5–6 and costs an afternoon, so it goes first regardless of enthusiasm. Step 3 is deliberately placed before step 5: it is cheap, it is self-contained, and it is useful to have one thing land before starting the one that might not.

Direction B is the fallback rather than the plan, because it survives contact with a negative result: if Hazard 3.2 closes Direction A, nothing about Direction B changes.

8 Priors

Stated before the work, so they can be scored afterwards.

	claim	prior	if it lands
2.1	the shape/position asymmetry is a paving	70%	reframes `q-container`
5.1	q-containers are $\mathbb{F}_q$-paved spaces, on a subcategory	60%	central theorem for `q-container`
6	Grassmannian container has series $\begin{bmatrix} n \\ k \end{bmatrix}_q$	90%	Targets row, free
1	the Hecke span is a q-bicomodule	75%	definition is on the right track
2	composition gives $(q-1)T_i + q$	45%	the definition is correct
3	RSK is a q-bicomodule	35%	the two repositories join
—	Direction A yields something publishable	25–35%	a container-native Hecke theory
3.2	the distributive law is the obstruction	50%	informative either way
4.1	expander codes are container morphisms	80%	applied write-up
4	the rate bound falls out of rank monotonicity	55%	the formalism earns its keep
—	anything here reaches arithmetic Langlands	$< 5\%$	—

The honest summary of that table: the highest-probability rows are the ones that improve documents you have already written, and the highest-ceiling row is at 25–35% and gated on a paper you cannot currently read. That is a normal shape for a research programme and is not an argument against it. It is an argument for doing step 1 this week.

9 Sources to obtain

LICS 2024, no-go theorems for directed-container distributive laws.
Paywalled; listed unobtained in `lascoux/papers/INDEX.md`. Gates Direction A. Highest priority.

Ahman, Uustalu.
Directed containers as categories. The comonoid half of Definition 3.1.

Garner.
Polynomial comonads / the comonoid–category correspondence, and bicomodules as the layer above.

Spivak, Niu.
Polynomial Functors: A Mathematical Theory of Interaction. Bicomodules in **Poly**; the ambient calculus for Definition 3.1.

Macdonald.
Symmetric Functions and Hall Polynomials, Ch. V. Step 2 of §7.

Sipser, Spielman.
Expander codes, IEEE Trans. Inform. Theory 42 (1996). Direction B.

Lubotzky, Phillips, Sarnak.
Ramanujan graphs, Combinatorica 8 (1988). Milestone 5.